Analytic Epidemiology - Study Designs I

APHS 20203 • Epidemiology & Biostatistics

Epidemiological Study: Basic Question
Are exposure and outcome associated?
More specifically:

- Do people with different exposures experience different health outcomes?
- Is the observed association likely to be causal, or could it reflect other explanations?

Classification of Study Designs

Two Categories of Analytic Studies

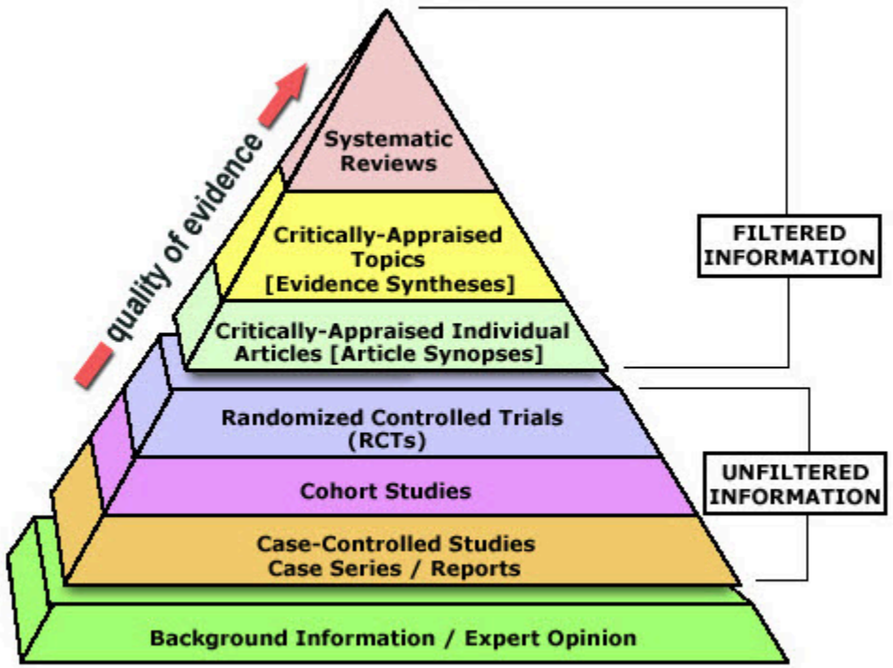
Observational design

- Investigator does not assign exposure
- Observes existing differences in exposure and outcome
- Primary goal: identify associations
- Examples include ecologic, cohort, and case-control studies

Experimental design

- Investigator assigns exposure or intervention
- Often uses randomization
- Primary goal: evaluate causal effects
- Examples include randomized controlled trials (RCTs)

Levels of Evidence in Epidemiologic Research



Generate Hypotheses
Establish Causality

- Study designs vary in how strongly they support causal inference
- Lower levels are often used to generate hypotheses
- Higher levels are better suited to test causal effects
- Moving up the pyramid generally means:
  - Greater control over alternative explanations
  - Stronger ability to evaluate causality

Important note:
Higher evidence does not mean “always correct,” and lower evidence does not mean “wrong.”

Epidemiologic Inference: From Association to Conclusion

Analytic epidemiology asks whether an observed association reflects a causal relationship.

- Step 1: Identify an association: Do exposure and outcome occur together more often than expected?
- Step 2: Evaluate validity:
  - Internal validity: Is the association likely to be real within this study?
  - External validity: Can the findings be generalized to other populations?
- Step 3: Consider alternative explanations
  - Could the association reflect chance?
  - Could it reflect systematic differences in how groups are compared or observed?
  - Could it reflect other factors related to both exposure and outcome?

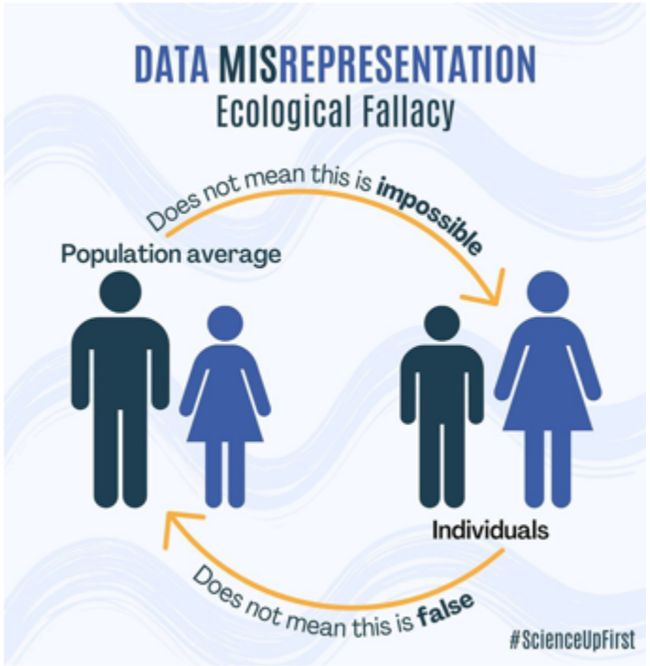
Key takeaway: Study design and validity assessment determine how confidently we can move from association to inference.

Ecologic Studies: Limits of Causal Inference

- Ecologic Studies: Group-level analysis
- Unit of analysis: groups (not individuals)
- Exposure & outcome: measured at the population level
- Common use: explore population patterns; generate hypotheses
- Key caution: group-level associations ≠ individual-level conclusions
- Take-home message: Ecologic studies are useful for population questions, but cannot establish individual-level causality.

Ecologic Fallacy

- Definition: Incorrectly applying group-level findings to individuals
  - Occurs when associations observed in populations are assumed to hold at the individual level
- Example
  - Observation: Cities with higher income have higher CHD rates
  - Incorrect inference: Wealthier individuals have higher CHD risk
  - Reality: Within cities, CHD risk may be higher among lower-income individuals
- Take-home message: Group-level data cannot be used to draw individual-level conclusions.



https://www.scienceupfirst.com/project/data-misrepresentation-ecological-fallacy/

Ecologic Study: Strengths & Limitations

Strengths	Limitations
• Uses existing population-level data	• Ecologic fallacy
• Efficient: fast and low cost	• Crude exposure measurement
• Useful when individual-level data are unavailable	• Limited control of confounding

Discussion: Ecologic Study Example

- Study
  - Sasaki et al. (1993): dietary fat intake & breast cancer mortality
  - Unit of analysis: countries (30 countries total)
- Key finding
  - Higher animal fat intake was associated with higher breast cancer mortality
  - Fish fat intake was associated with lower breast cancer mortality
- Discussion prompts
  - What makes this study feasible and efficient?
  - Why can these results not be interpreted at the individual level?
  - What other population-level factors could explain the association?

Sasaki S, Horacsek M, Kesteloot H. An ecological study of the relationship between dietary fat intake and breast cancer mortality. Prev Med. 1993 Mar;22(2):187-202. doi: 10.1006/pmed.1993.1016. PMID: 8483858.

Experimental Studies

- Key feature: investigator assigns exposure or intervention
  - Assigning exposure allows stronger conclusions about cause and effect.
- Comparison: intervention group vs control group
- Goal: evaluate the effect of an intervention
- Two broad types
  - True experimental studies: include randomization
  - Quasi-experimental studies: no randomization, but intervention is assigned
- Why they matter
  - Stronger control of alternative explanations
  - Better support for causal inference than observational designs

Intervention Studies

- Key Features
  - A type of experimental study
  - Introduce a planned action to assess its impact
  - Conducted in real-world settings
- Common examples
  - Medical interventions (e.g., vaccines, treatments)
  - Behavioral programs (e.g., smoking cessation)
  - Policy or community interventions (e.g., taxation, education)
- Take-home message: Intervention studies evaluate whether deliberate actions change health outcomes in practice.

Experimental vs Intervention Studies

- Experimental studies
  - Broad category where the investigator manipulates a condition
  - Often conducted in controlled settings
  - Example: laboratory study testing how temperature affects a biological process
- Intervention studies
  - A type of experimental study focused on planned actions to improve health
  - Conducted in real-world populations
  - Example: introducing a vaccination program or a smoking cessation policy
- Key point: All intervention studies are experimental, but not all experimental studies are health interventions.

Exercise: Experimental vs Intervention Studies

- Exercise: For each example below, decide whether it is an Experimental study or an Intervention study.
- Researchers expose bacteria to different temperatures in a lab and measure growth rate.
- A city introduces a new bike-lane policy to encourage physical activity and reduce injuries.
- Participants are randomly assigned to receive a new medication or a placebo to evaluate treatment effectiveness.
- Psychologists test whether blue or red lighting affects concentration in a controlled setting.
- Hint
  - Which examples involve a planned action to improve health?
  - Which examples involve manipulation without a health intervention?

Clinical Trials: Medical Intervention Studies

- Clinical Trials
  - A type of intervention study conducted in humans
  - Participants receive a medical intervention under controlled conditions
  - Designed to evaluate safety and effectiveness
- Common purposes
  - Prevention (prophylactic trials, e.g., vaccines)
  - Treatment (therapeutic trials, e.g., new drugs or procedures)
- Key takeaway: Clinical trials provide strong evidence for whether medical interventions work and are safe.

Randomized Controlled Trials (RCTs)

- Randomized Controlled Trials (RCTs)
  - A type of clinical trial
  - Participants are randomly assigned to intervention or control groups
  - Groups are comparable at baseline by design
  - Outcomes are measured consistently across groups
- Why randomization matters
  - Reduces systematic differences between groups
  - Strengthens causal inference
- Take-home message: Randomization helps isolate the effect of the intervention, making RCTs the strongest design for evaluating causality.

RCT: Randomization

- Randomization
  - Assigns participants to groups by chance
  - Balances known and unknown factors across groups
  - Occurs before outcomes are measured
- What randomization does
  - Reduces systematic differences between groups
  - Minimizes alternative explanations for observed effects

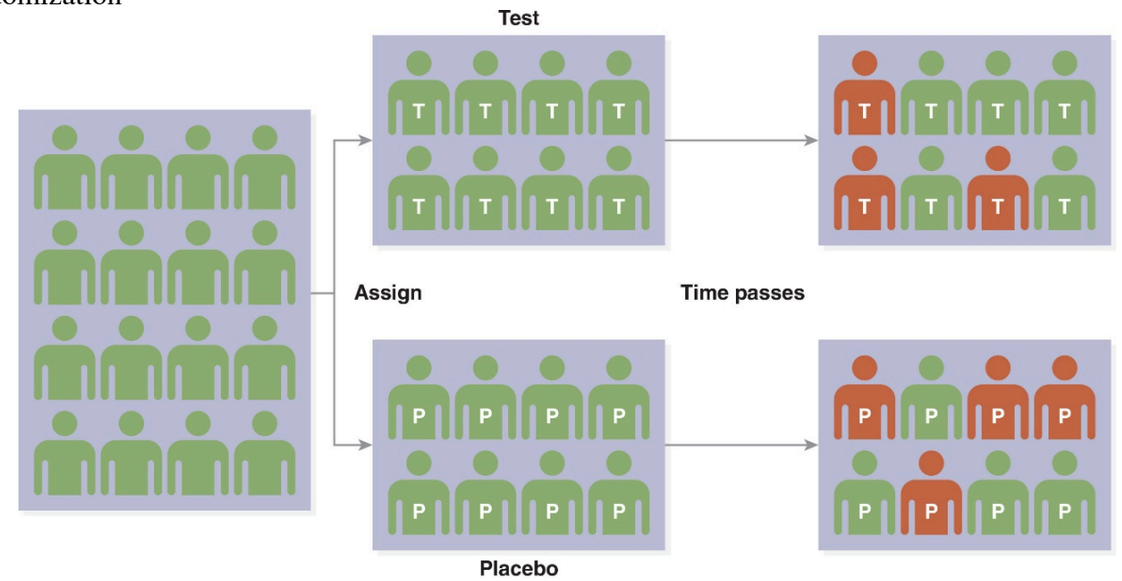


Figure 7.5 Diagram of a randomized controlled trial

Indication that red means occurrence of the outcome.

Take-home message

Randomization improves internal validity by making groups comparable at baseline.

Discussion: Randomized Controlled Trial (RCT) Example

- Study overview
  - Large randomized controlled trial of a low-fat dietary pattern
  - Postmenopausal women assigned to intervention or usual diet
  - Long-term follow-up
- Discussion prompts
  - Why is an RCT an appropriate design for this question?
  - What features of this study strengthen causal inference?
  - What challenges might arise in long-term behavioral interventions?
- Key takeaway: RCTs strengthen causal conclusions, but feasibility, adherence, and follow-up matter.

Prentice RL, Caan B, Chlebowski RT, et al. Low-Fat Dietary Pattern and Risk of Invasive Breast Cancer: The Women's Health Initiative Randomized Controlled Dietary Modification Trial. JAMA. 2006;295(6):629–642. doi:10.1001/jama.295.6.629

Discussion: Comparing Study Designs

- Comparing Two Studies: Think back to the two studies discussed today:
  - Ecologic study: dietary fat intake and breast cancer mortality
  - Randomized controlled trial: low-fat dietary intervention
- Discussion questions
  - What is the unit of analysis in each study?
  - Which study provides stronger evidence for causality, and why?
  - What kinds of conclusions are appropriate for each design?

RCTs: Strengths & Limitations

Strengths

- Randomization improves group comparability
- Strong internal validity
- Best design for evaluating causal effects of interventions

Limitations

- Often expensive and time-consuming
- May have limited generalizability to real-world populations
- Not always ethical or feasible

Project Prep 2

Revisit: Study Design & Causal Inference

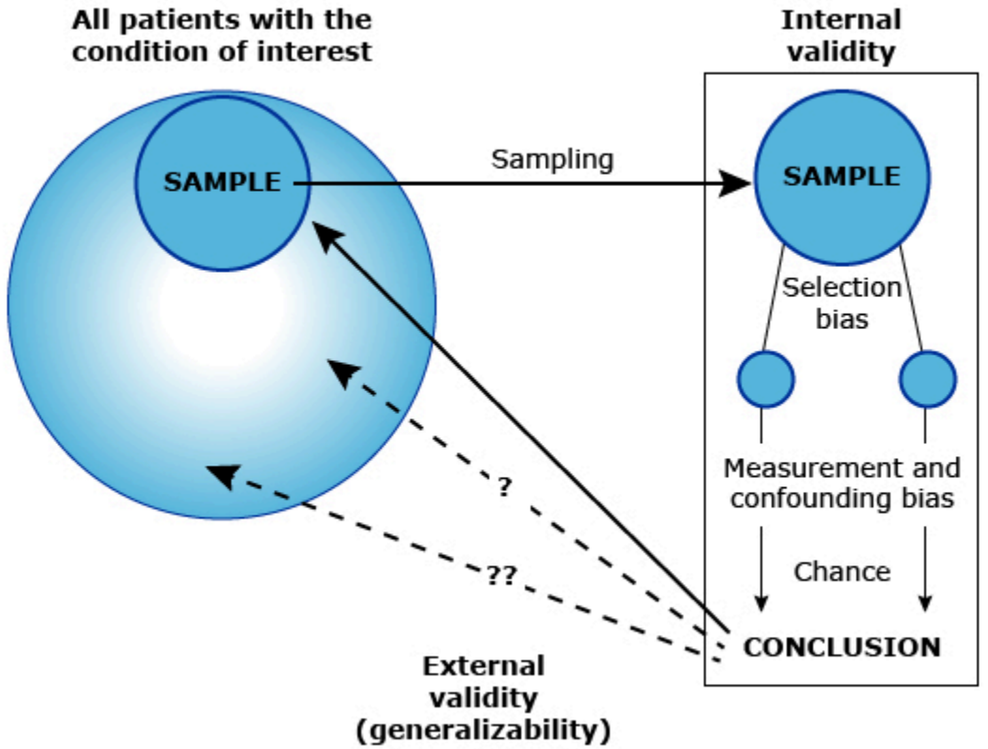
- Review:
  - Ecologic study: dietary fat & breast cancer mortality
  - RCT: low-fat dietary intervention
- Quick question
  - Which design offers stronger evidence for causality?
- Follow-up
  - What might still limit confidence in either study?
- Key idea: Even strong study designs have limitations. That leads us to validity.

Challenges to the Validity of Study Designs

- Even well-designed studies can produce misleading results.
- Two central questions:
  - External validity: Can the findings be generalized to other populations or settings?
  - Internal validity: Is the observed association likely to reflect a true causal relationship?
- If validity is compromised, conclusions may reflect:
  - Random error (chance)
  - Systematic error
  - Confounding
- Key idea: Strong design does not eliminate threats to validity.

External Validity

- Definition: The extent to which study results can be generalized to other populations or settings
- Key influences
  - Sampling error: random variation
  - Sampling bias: systematic lack of representativeness
- Guiding question: Do these results apply beyond this sample?



Sampling Error

- Definition: Random variation that occurs because we study a sample, not the entire population.
- Key points
  - Occurs whenever a sample is drawn
  - Affects the precision of estimates
  - Not a design flaw — a natural result of sampling
- How to reduce it: Increase sample size
- Example
  - If the true population smoking rate is 20%,
  - A sample of 50 people might estimate 16% or 24% due to chance.
- Key takeaway: Sampling error reflects random variation, not systematic bias.

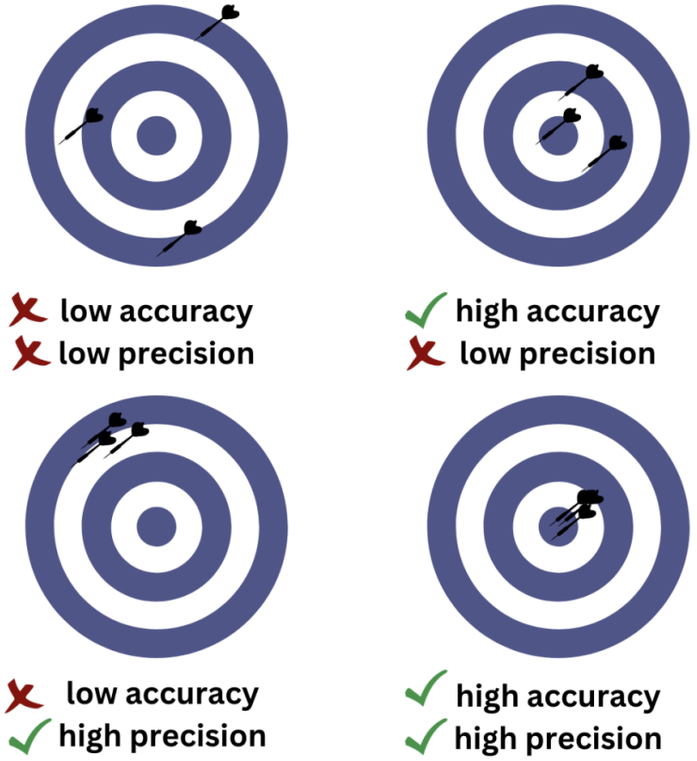
Sampling Bias

- Definition: A systematic error that occurs when the sample is not representative of the target population.
- Key points
  - Results from how participants are selected
  - Affects both generalizability and potentially conclusions
  - Unlike sampling error, it does not disappear with larger sample size
- Example
  - Studying hypertension using patients from a single specialty clinic may overrepresent more severe cases.
- Key takeaway: Sampling bias threatens external validity because the sample does not reflect the population of interest.

Internal Validity

Random Error vs Systematic Error

- Random Error (Chance)
  - Natural variability in data
  - Directly affects precision (spread of estimates)
  - Reduced by increasing sample size
- Systematic Error (Bias)
  - Flaws in design, selection, or measurement
  - Affects accuracy (closeness to the truth)
  - Not corrected by larger sample size



Exercise: Random or Systematic?

For each scenario, decide: Random error (chance) or Systematic error (bias)

- A random sample of 40 students is used to estimate smoking prevalence on campus. The study finds 16% prevalence, while the true campus rate is 20%.
- In a survey about alcohol use, participants are asked face-to-face by a researcher. Many participants report lower drinking levels than expected.
- In a cohort study of pesticide exposure and Parkinson's disease, more participants in the exposed group drop out during follow-up than in the unexposed group.
- Blood pressure is measured using two different devices. One device consistently gives readings 5 mmHg higher than the other.

Bias (Systematic Error)

- Definition: A systematic error in the design, conduct, or measurement of a study that distorts the true association.
- Key features
  - Shifts results in a consistent direction
  - Affects accuracy, not precision
  - Not corrected by increasing sample size
- Two major categories
  - Selection bias
  - Information bias
- Key idea: Bias creates a misleading estimate of the exposure–outcome relationship.

Selection Bias

- Definition: A systematic error caused by differences in who enters or remains in the study.
- How it occurs
  - Differences in participant selection
  - Differences in participation or refusal
  - Differences in loss to follow-up
- Key idea: When the groups being compared differ because of the selection or retention process, the association may be distorted.

Common Types of Selection Bias

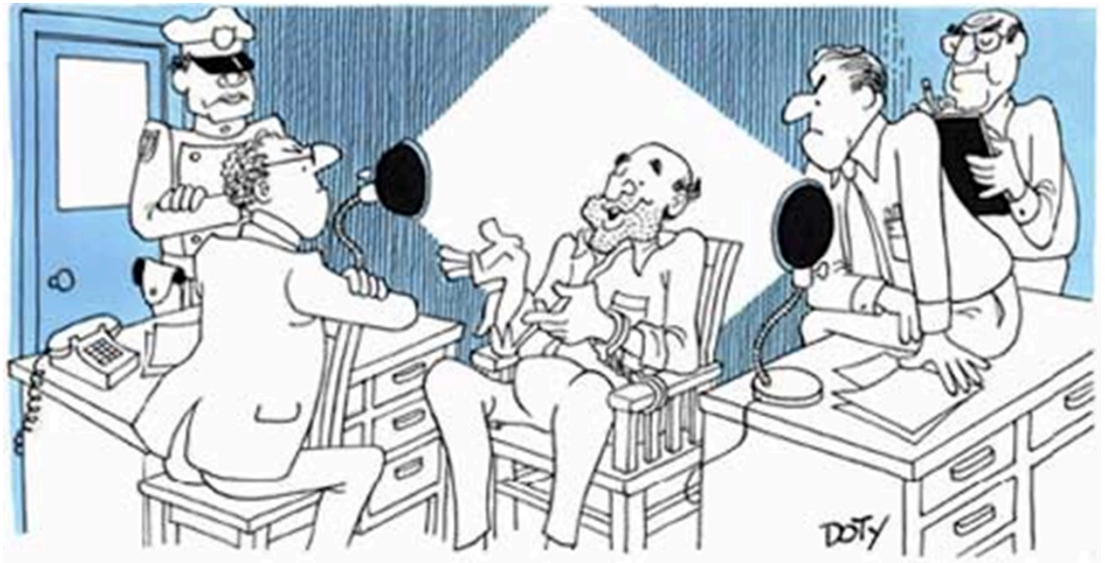
- Self-selection bias: Participation or refusal is related to exposure or outcome
- Healthy worker effect: Workers tend to be healthier than the general population
- Control selection bias: In case-control studies, controls do not represent the source population
- Loss to follow-up (attrition bias): In cohort studies, dropout differs between exposure groups
- Key idea: Selection bias occurs when group differences arise from who enters or remains in the study.

Information Bias

- Definition
  A systematic error caused by inaccurate or inconsistent measurement of exposure or outcome.
- How it occurs
  - Differences in how information is reported
  - Differences in how information is collected
  - Differences in how information is recorded
- Key idea: When exposure or outcome data differ between groups, the association may be distorted.

Types of Information Bias: Recall Bias

- Definition: A type of information bias in which participants remember past exposures differently.
- When it occurs
  - Often in case-control studies
  - Cases may recall exposures more carefully than controls
- Key idea: Differences in memory between groups can distort the exposure–outcome association.



Types of Information Bias: Researcher Bias

- Interviewer Bias
  - Occurs during data collection
  - Interviewer knowledge influences how questions are asked
  - Can lead to different responses between groups
- Observer Bias
  - Occurs during outcome assessment
  - Observer expectations influence how outcomes are recorded
  - More likely when blinding is absent
- Key idea: When researchers influence data collection or measurement, results may be systematically distorted.



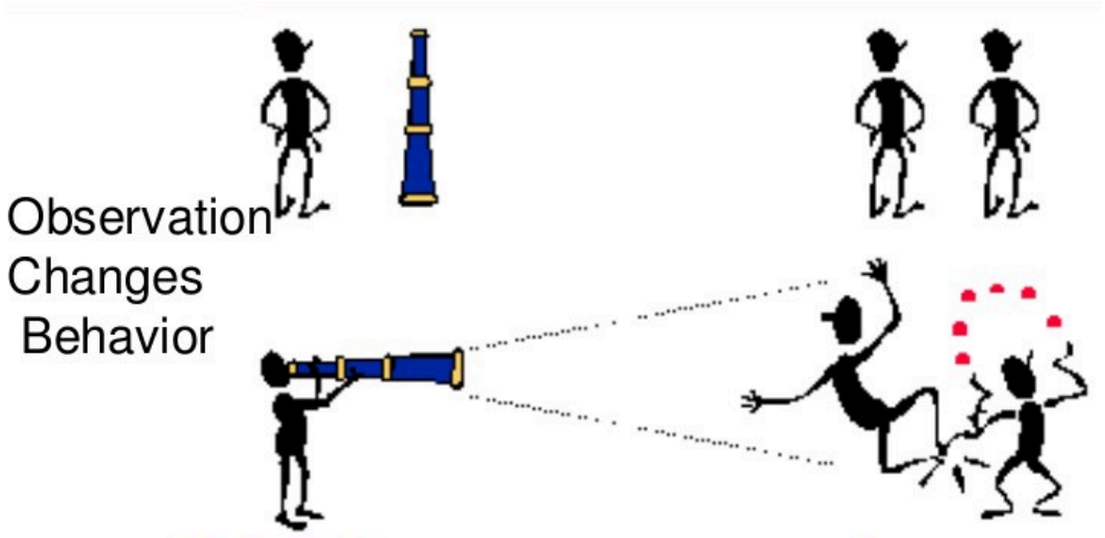
Types of Information Bias: Social Desirability

- Definition: participants provide responses that are viewed as socially acceptable rather than accurate.
- When it occurs
  - Sensitive behaviors (e.g., smoking, alcohol use, drug use)
  - Self-reported health behaviors or lifestyle factors
- Key idea: When participants modify responses to appear favorable, exposure or outcome data may be systematically distorted.


QUESTIONNAIRE
DOTY

Types of Information Bias: Hawthorne effect

- Definition: participants change behavior because they know they are being observed.
- When it occurs
  - During clinical trials or behavioral studies
  - When participants are monitored or aware of measurement
- Key idea: when behavior changes due to observation, the measured exposure or outcome may not reflect usual behavior.



Exercise: Identify the Type of Information Bias

For each scenario, identify the most likely type of information bias and explain why.

- In a case-control study, parents of sick children recall pesticide exposure during pregnancy more carefully than parents of healthy children.
- Participants report lower alcohol use in face-to-face interviews than expected from sales data.
- In a clinical trial, outcome assessors know which participants received the treatment and rate them more favorably.
- Interviewers ask more detailed exposure questions to lung cancer cases than to controls.
- Participants increase exercise levels during a study because they know their activity is being monitored.

Reducing Selection and Information Bias

- Reducing Selection Bias
  - Use appropriate comparison groups
  - Apply clear and consistent inclusion criteria
  - Minimize loss to follow-up
  - Ensure recruitment is not related to both exposure and outcome
- Reducing Information Bias
  - Use standardized definitions and procedures
  - Apply blinding when possible
  - Collect data consistently across groups
  - Use validated measurement tools
- Key idea: Careful design and consistent measurement strengthen internal validity.

Confounding

- Definition: a distortion of the exposure–outcome association caused by a third variable.
- How to identify a confounder
  - Associated with the exposure
  - Independently associated with the outcome
  - Not caused by the exposure
- Example
  - Smoking may confound the association between alcohol use and lung cancer.
- Key idea: If a third variable meets these criteria, the observed association may not reflect a true causal relationship.

Scenario

- Studying the effect of Alcohol Use (exposure) on Lung Cancer (outcome).

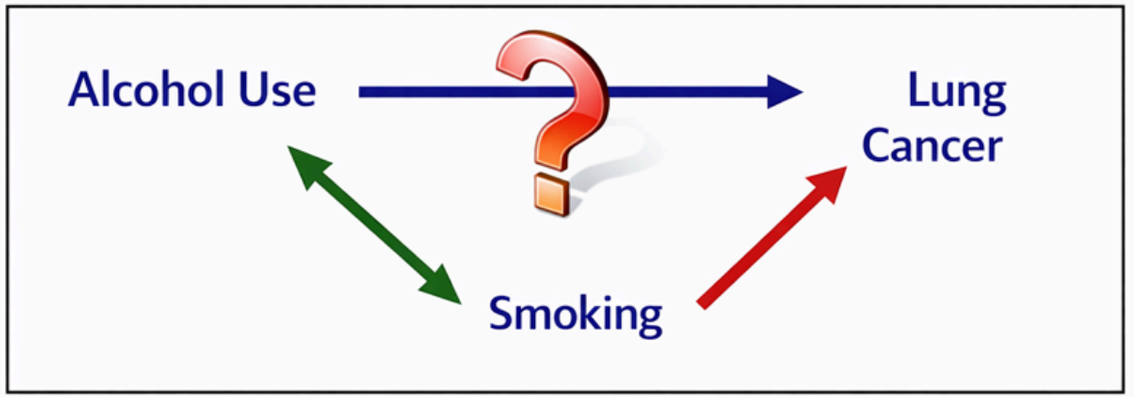
Confounder: Smoking

- Associated with alcohol use
- Independently associated with lung cancer
- Not caused by alcohol use

Effect

- The observed association between alcohol and lung cancer may be distorted because smoking is related to both.

Example of Confounding



✓ Associated with the exposure (Alcohol use)

✓ Independently associated with the outcome (Lung cancer)

✓ Not on the causal pathway

Exercise: Could This Be Confounding?

For each scenario, decide whether the association could be explained by confounding. Identify a possible confounder and explain why.

- Scenario 1: A study finds that people who carry lighters have a higher risk of lung cancer.
  - What third variable might explain this association?
  - Does it meet the criteria for a confounder?
- Scenario 2: A study finds that children who sleep with the light on are more likely to develop myopia (nearsightedness).
  - What third variable might explain this association?
  - Does it meet the criteria for a confounder?

Addressing Confounding

- Confounding cannot be ignored when interpreting results.
  - Randomization helps balance confounders between groups
  - In observational studies, researchers must identify and measure potential confounders
  - Results should be interpreted cautiously if important confounders are not accounted for
- Key idea: If confounding is not addressed, an observed association may not reflect a true causal relationship.

Wrap-Up

Causal inference depends on:

- Study design
  - Observational vs experimental
  - Ecologic vs RCT
- Validity
  - External validity
  - Internal validity

Threats to internal validity

- Chance
- Selection bias
- Information bias
- Confounding

Key message: Strong conclusions require both good design and strong validity.

Final Epi Presentation
Get started!